

Annexure – 1

Proposed Solution Template (Open Challenge)

- 1. Applicant Name:** Ayush Srivastava & Aditya Srivastava
- 2. Startup/MSME Name:** BharatVigil
- 3. Challenge Title:** Geospatial Intelligence Platform for Border Surveillance and Strategic Infrastructure Monitoring
- 4. Proposed Duration:** 12 months
- 5. Contact & Email ID:** +91 9905014104, ayusrivastava2006@gmail.com
+91 7709178983 adi.srivastav02@gmail.com

1. Brief Summary of the Proposed Solution (≤250 words)

India's land borders span over 15,000 km, much of it high-altitude, remote terrain where ground patrols are infrequent and weather-limited. Adversary infrastructure build-up — roads, bridges, helipads, forward camps — is currently tracked through manual reconnaissance and ad-hoc satellite tasking, creating intelligence gaps of weeks to months.

BharatVigil is an AI-driven geospatial intelligence platform that continuously monitors border regions using multi-temporal SAR and optical satellite imagery. Deep-learning-based change detection identifies new construction and infrastructure expansion, assigning each change a confidence and threat-severity score, surfaced through a Command Intelligence Console with automated alert routing and military-formatted intelligence briefs.

The data layer is sensor-agnostic by design: open-access Sentinel-1/2 imagery accelerates AI model development today, while the architecture is built to migrate onto India's own indigenous SAR constellation — ISRO's RISAT/EOS-04, NISAR, and Cartosat — for live, classified operations, since open foreign-controlled data cannot serve tactical-level defence tasking. The system is cloud-native today, with a roadmap to on-premise/air-gapped deployment for full data sovereignty.

Our niche is deliberate: we are not competing with Palantir on raw analytics power or Maxar on imagery resolution — we are the indigenous, sovereign, field-deployable alternative built specifically for Indian terrain, reporting workflows, and classified-data constraints that foreign platforms structurally cannot serve. This reduces the time between a ground change occurring and a commander knowing about it from weeks to hours.

(228 words)

2. Key Technology(s) Used (5–6 keywords)

1. SAR & Optical Remote Sensing
2. Deep Learning Change Detection
3. Geospatial AI / GIS
4. Threat Intelligence Scoring
5. Cloud-Native / Sovereign Deployment Architecture
6. Multi-Sensor Data Fusion

3. Deliverable(s)

S. No	Deliverable Name	Brief Description
1	Multi-Sensor Ingestion & Preprocessing Pipeline	Sensor-agnostic Sentinel-1/2 ingestion for development, with adapter design ready to migrate to ISRO RISAT/EOS-04/NISAR feeds; SAR despeckling, calibration, and geo-rectification module
2	AI Change Intelligence Engine	Deep-learning model for multi-temporal change detection, classification, and threat-severity scoring
3	Command Intelligence Console	Web-based GEOINT dashboard with layered map intelligence, AOI monitoring, and alert routing
4	Automated Intelligence Brief Generator	Templated, exportable reports formatted for military reporting workflows
5	Pilot Deployment Package	Tested deployment on a designated AOI with on-prem deployment proof-of-concept

4. Proposed Timeline(s) (Phase-wise, in months)

Phase	Duration	Milestone
Phase 1: Foundation	Month 0–3	End-to-end pipeline on historical Sentinel-1/2 data for 1–2 AOIs (open-access, for rapid model development); baseline change-detection model with accuracy benchmarking
Phase 2: Defence-Tuning	Month 3–6	SAR backscatter change-detection module; threat-severity scoring refined with domain-expert input; automated intelligence-brief generation
Phase 3: Pilot Readiness	Month 6–9	Security hardening (RBAC, audit logging), on-prem deployment proof-of-concept, NRSC/ISRO data-sharing engagement to onboard indigenous RISAT/EOS-04/NISAR feeds, partner-agency engagement
Phase 4: Pilot Deployment	Month 9–12	Live pilot with partner unit on a real AOI; feedback-driven false-positive reduction

Annexure – 2

BharatVigil — Proposed Technical Solution (Detailed)

Geospatial Intelligence Platform for Border Surveillance and Strategic Infrastructure Monitoring

1. Technical Architecture & Approach

BharatVigil follows a five-layer pipeline architecture that takes raw satellite imagery and converts it into commander-ready intelligence within hours of a ground change occurring. Each layer is loosely coupled and sensor-agnostic, so the imagery source can be swapped (open-access, ISRO, or commercial) without re-architecting the system — a deliberate design choice explained further in Section 4.

1. Data Ingestion Layer

Automated tasking and download of SAR and optical imagery for defined Areas of Interest (AOIs); supports multiple satellite sources through a common adapter interface.

2. Preprocessing Layer

SAR despeckling, radiometric calibration, orthorectification, and co-registration of multi-temporal, multi-sensor imagery onto a common geospatial grid.

3. AI Change Intelligence Engine

Deep-learning models (Siamese CNN / transformer-based change detection) compare current vs. historical imagery to flag new construction, road formation, vehicle staging, and terrain alteration, each tagged with a confidence score.

4. Threat Scoring & Fusion Layer

Rule-based and ML-assisted severity scoring that weighs change type, proximity to the Line of Control/Line of Actual Control, rate of change, and historical pattern-of-life, fusing SAR and optical detections into a single verified alert.

5. Command Intelligence Console

Web-based GEOINT dashboard with layered maps, AOI monitoring, alert routing to designated units, and one-click export into military-formatted intelligence briefs.

2. Innovation

Sensor-agnostic ingestion

the same change-detection engine works across data sources, allowing a seamless transition from open-access imagery used for prototyping to sovereign Indian satellite data for operational deployment.

Threat-severity scoring (not just change detection)

most civilian change-detection tools stop at “something changed here”; BharatVigil adds a defence-specific severity model trained with domain-expert input from retired military geospatial analysts.

Military-format automated briefs

detections are auto-compiled into standard intelligence-brief templates, removing the manual report-writing step that currently delays commander awareness.

Air-gapped deployment roadmap

architecture is containerised from day one so the entire stack can be deployed on isolated, on-premise infrastructure without any external network dependency — a core requirement for classified environments.

3. Implementation & Feasibility

Development follows the four-phase plan in Annexure-1 (Foundation → Defence-Tuning → Pilot Readiness → Pilot Deployment). Feasibility is anchored in three facts: Sentinel-1/2 imagery is freely available today for rapid model development without procurement delays; the underlying deep-learning architectures (Siamese networks, U-Net-style segmentation) are well-proven in academic and commercial change-detection literature; and the system is cloud-native using standard containerised microservices, making both scaling out (more AOIs) and migration to on-premise/air-gapped infrastructure straightforward.

Technology Stack

Component	Approach
Data sources	Sentinel-1 (SAR), Sentinel-2 (optical) for development; ISRO RISAT/EOS-04, Cartosat, NISAR for operational deployment (see Section 4)
Processing & AI	Python, PyTorch/TensorFlow, GDAL/SNAP for SAR preprocessing, containerised inference services
Platform	Kubernetes-based microservices; cloud-native for development, redeployable on-premise/air-gapped for production
Console	Web-based GIS dashboard (React + open-source mapping stack such as Leaflet/Mapbox GL) with RBAC and audit logging

Scalability is handled at two levels: horizontally, by adding AOIs and satellite passes as parallel processing jobs on the same containerised pipeline; and organisationally, by onboarding additional partner units onto the Command Intelligence Console with role-based access, so a single backend can serve multiple commands without re-deployment.

4. Challenges & Mitigation

The most material technical risk for this proposal is the data source itself, and it deserves direct treatment.

Is Sentinel-1/2 imagery acceptable for Indian Army / classified use?

Short answer: not for live operational or classified tasking, only for development, training, and proof-of-concept. Sentinel-1 and Sentinel-2 are part of the European Space Agency's Copernicus programme — free, open-access data that is openly published and controlled by a foreign agency. That makes it valuable for rapid model development (no procurement cycle, large historical archive, global coverage) but unsuitable as the primary feed for defence-grade surveillance for three structural reasons:

- **Sovereignty & continuity** — tasking, revisit timing, and continued access are controlled by ESA, not India; there is no guarantee of priority access or continuity during a crisis, and the acquisition cannot be tied to a classified data-handling chain.
- **Resolution** — Sentinel-1 SAR (~5–20 m) and Sentinel-2 optical (10–60 m) are adequate for detecting large infrastructure change but fall short of the sub-metre to few-metre resolution Indian forces use for tactical-level intelligence on individual structures, vehicles, or troop staging.
- **Open-access nature** — because the imagery is public, an adversary has access to the exact same raw data, which limits its value as a unique intelligence advantage in a contested border context.

This is exactly why the proposal's core sovereignty argument (Annexure-1) matters operationally, not just commercially. India already has — and is rapidly expanding — an indigenous all-weather SAR constellation that BharatVigil is designed to migrate onto:

Indigenous Source	Relevance to BharatVigil
ISRO RISAT / EOS-04 series	Indigenous C-band SAR (EOS-04/RISAT-1A) already operational; resolution down to ~1 m in spotlight mode, purpose-built for border surveillance and used in past counter-infiltration operations.
NISAR (ISRO–NASA)	Launched 2025; dual L/S-band SAR providing global coverage every 12 days with centimetre-scale deformation sensitivity, useful for detecting subsurface/tunnel activity along the LAC.
Cartosat series	High-resolution panchromatic optical imagery used operationally for Indian defence mapping and target-area imaging.
Space-Based Surveillance Phase-III	Government-approved programme (₹26,968 crore) for 52 dedicated surveillance satellites for round-the-clock, all-weather reconnaissance, with ISRO and Indian private players jointly building the constellation.
Private Indian NewSpace (e.g. GalaxEye)	Indian start-ups are now launching indigenous OptoSAR (combined optical+SAR) satellites aimed explicitly at building a sovereign, high-frequency Earth observation network for defence and commercial users.

Mitigation strategy: BharatVigil's data-ingestion layer is built as a source-agnostic adapter from day one (Section 1). Phase 1–2 (Months 0–6) use Sentinel-1/2 purely to build, train, and validate the change-detection and threat-scoring models against historical, open-access imagery — this is standard practice even for classified systems, since model development does not require classified inputs. From Phase 3 onward (Month 6+), the same pipeline is reconfigured to ingest ISRO RISAT/EOS-04, Cartosat, and eventually NISAR data through the National Remote Sensing Centre (NRSC) / partner-agency data-sharing channel, with on-premise or air-gapped deployment ensuring the classified feed never touches external infrastructure. This phased approach also de-risks the proposal: it proves the AI engine's accuracy cheaply and quickly on open data before any classified data-sharing agreement is even needed.

Other Technical Risks

- **False positives at scale** — Natural seasonal changes (snowmelt, vegetation, riverbed shifts) can mimic construction activity in SAR imagery. Mitigated via a multi-season baseline model and human-in-the-loop verification before an alert is escalated to a brief.
- **Cloud cover (optical only)** — Optical (Sentinel-2/Cartosat) imagery is unusable under persistent cloud cover in Himalayan border regions. Mitigated by prioritising SAR as the primary all-weather feed and using optical only for confirmation/visual context.
- **Data-sharing & clearances** — Transition to ISRO/NRSC defence-grade feeds requires formal data-sharing agreements and security clearances. Mitigated by engaging NRSC/partner-agency stakeholders from Phase 3 (Month 6) onward, well before the Month 9 pilot, as already reflected in the Annexure-1 timeline.

5. Visuals & Supporting Data

A layered architecture diagram and the data-source comparison table above are intended to accompany this submission as supporting visuals. The recommended diagram flow for the pitch deck/demo is:

- Satellite Sources (Sentinel-1/2 → RISAT/EOS-04/NISAR roadmap) → Ingestion & Preprocessing
- Preprocessing → AI Change Intelligence Engine (multi-temporal comparison)
- Change Intelligence Engine → Threat Scoring & Fusion

- Threat Scoring & Fusion → Command Intelligence Console → Automated Intelligence Brief → Commander

6. Any Other Relevant Details

BharatVigil's phased data-sourcing approach — open-access imagery for rapid R&D, indigenous SAR for operational deployment — mirrors how DRDO and ISRO themselves have historically scaled defence space capability (e.g., RISAT-2 as a fast interim solution post-2008, followed by indigenous RISAT-1). This is offered as the project's standing answer to data-sovereignty questions in interview, and is reflected directly in the Phase 3–4 milestones of the Annexure-1 timeline.

Annexure – 3

BharatVigil — Advantages, Capabilities & Competencies

1. Technology Advantages (≤150 words)

- **All-weather, day/night coverage:** SAR-based change detection works through cloud cover, fog, and darkness — the dominant conditions along Himalayan and monsoon-affected border stretches where optical-only systems go blind for weeks.
- **Sensor-agnostic ingestion:** the same AI pipeline runs on open-access Sentinel-1/2 imagery today and on indigenous ISRO RISAT/EOS-04, NISAR, and Cartosat feeds tomorrow, with no re-architecture needed.
- **Defence-tuned change detection:** the deep-learning engine is trained to distinguish man-made infrastructure change (roads, helipads, forward camps) from natural seasonal variation, reducing false positives versus generic environmental-monitoring models.
- **Threat-severity scoring:** goes beyond “what changed” to “how much it matters,” combining change type, proximity to the LAC/LoC, and rate of change into one actionable score.
- **Cloud-native, air-gap-ready:** containerised microservices let the stack run in the cloud during development and redeploy on isolated, on-premise infrastructure for classified operations without rebuilding.

(134 words)

2. Product Advantages (≤150 words)

- **Hours, not weeks, to commander awareness:** automated detection, scoring, and military-formatted brief generation collapse a process that today depends on manual reconnaissance and ad-hoc satellite tasking.
- **Single console, multiple AOIs and units:** the Command Intelligence Console scales horizontally — new Areas of Interest and partner units are onboarded with role-based access, not separate deployments.
- **Built for the reporting workflow defence already uses:** outputs are exported as standard military intelligence briefs, not generic dashboards, minimising training overhead and adoption friction for end users.
- **Confidence-scored, human-verifiable alerts:** every flagged change carries a confidence score and supporting before/after imagery, keeping a human analyst in the loop before any alert is escalated.
- **Deployment flexibility:** the same product runs as a cloud pilot for fast evaluation and as an on-premise/air-gapped instance for operational rollout, matching the agency's security posture at each stage.

(137 words)

3. Commercial Advantages (≤150 words)

- **Lower cost than foreign GEOINT platforms:** no licensing dependency on Palantir-class analytics suites or Maxar-class commercial imagery contracts; the core engine runs on lower-cost open-access and ISRO-sourced imagery.
- **No foreign data-sovereignty exposure:** an indigenous, India-built and India-hosted platform avoids the legal and operational risk of routing classified border data through foreign-controlled infrastructure — a structural requirement foreign vendors cannot meet.

- **Single indigenous vendor across the pipeline:** ingestion, AI, console, and reporting are owned end-to-end, avoiding the integration cost and vendor lock-in of stitching together multiple foreign point solutions.
- **Phased, low-risk procurement path:** agencies can pilot on open-access data at near-zero data cost before committing to classified data-sharing agreements, reducing upfront commercial risk.
- **Reusable across users:** the same platform serves border-monitoring, infrastructure-protection, and disaster-response use cases, widening the addressable customer base beyond a single defence buyer.

(135 words)

4. Capabilities & Competencies (≤250 words)

- **Remote sensing & SAR processing:** hands-on experience with Sentinel-1/2 data pipelines, SAR despeckling, calibration, and multi-temporal geo-rectification.
- **Applied deep learning for Earth observation:** experience building and validating change-detection and image-classification models (CNN/transformer-based) on satellite imagery.
- **Full-stack platform development:** experience building cloud-native, containerised web applications and GIS dashboards (mapping stack, RBAC, alerting), suited to the Command Intelligence Console.
- **Geospatial domain knowledge:** working familiarity with GIS tooling (GDAL/QGIS-class workflows) and AOI-based monitoring relevant to border and infrastructure surveillance.
- **Understanding of defence reporting workflows:** design of the system's outputs around military-formatted intelligence-brief structures, informed by engagement with domain-expert input on threat-severity scoring.
- **Security-conscious system design:** architecture planned from the outset for RBAC, audit logging, and on-premise/air-gapped deployment, anticipating classified-environment requirements rather than retrofitting them.

(121 words)